Neural-network adaptive-order methods for high-fidelity flow simulations

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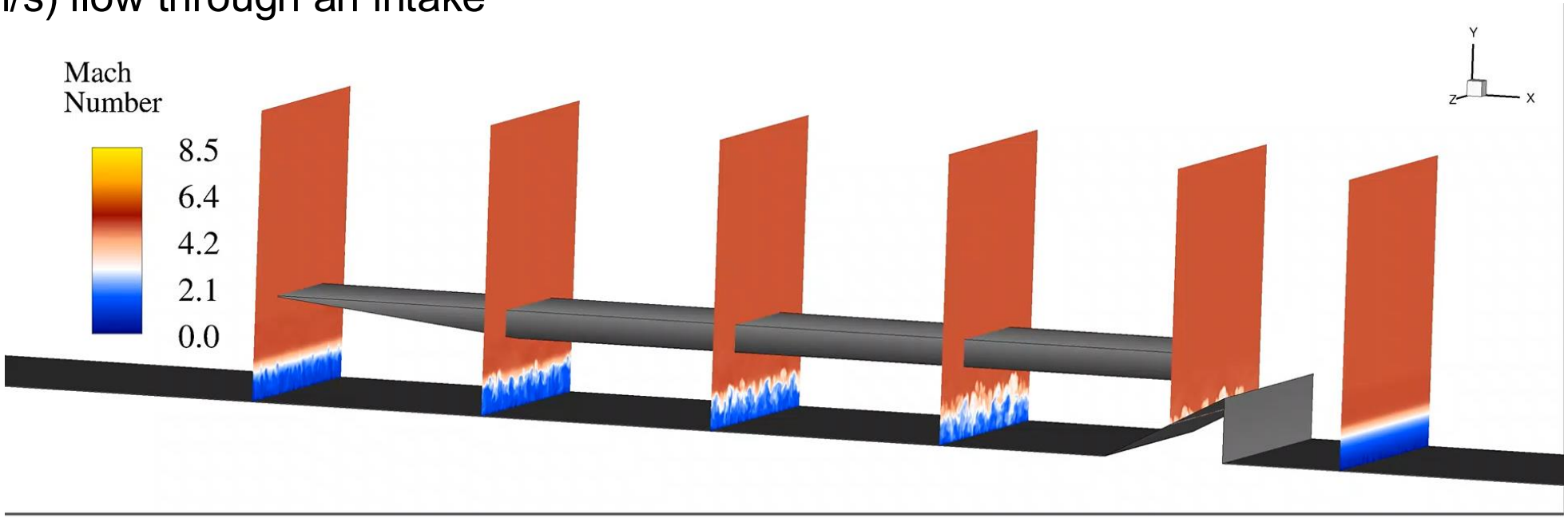
Mini-Visit Talk I-X, Imperial College London
June 25, 2026

Motivation

Aerodynamics: High-fidelity methods for simulating compressible flows (single or multiphase)

Mach 4.9 (~ 10^3 m/s) flow through an intake

Mach number contours



Density gradient contours

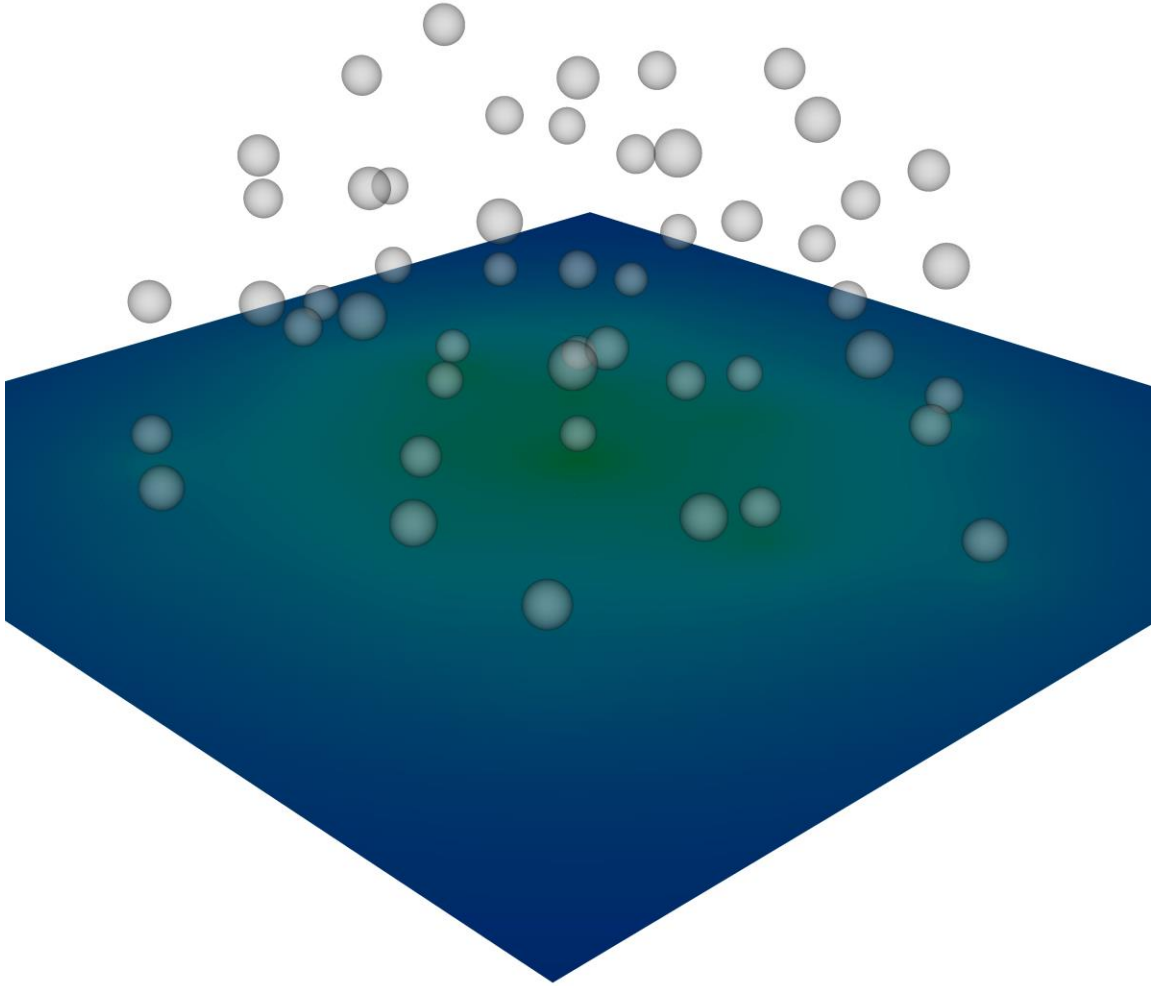


Motivation

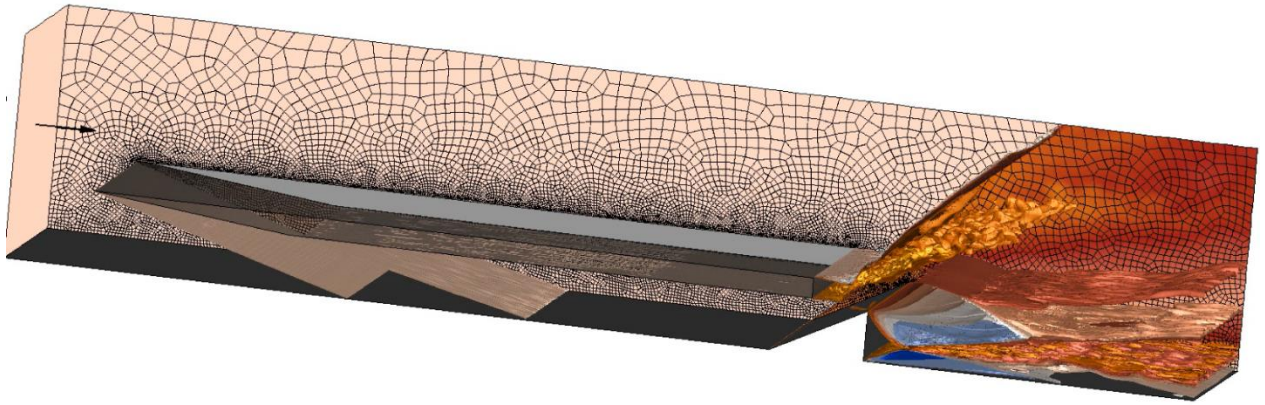
Hydrodynamics: High-fidelity methods for simulating cloud cavitation collapse

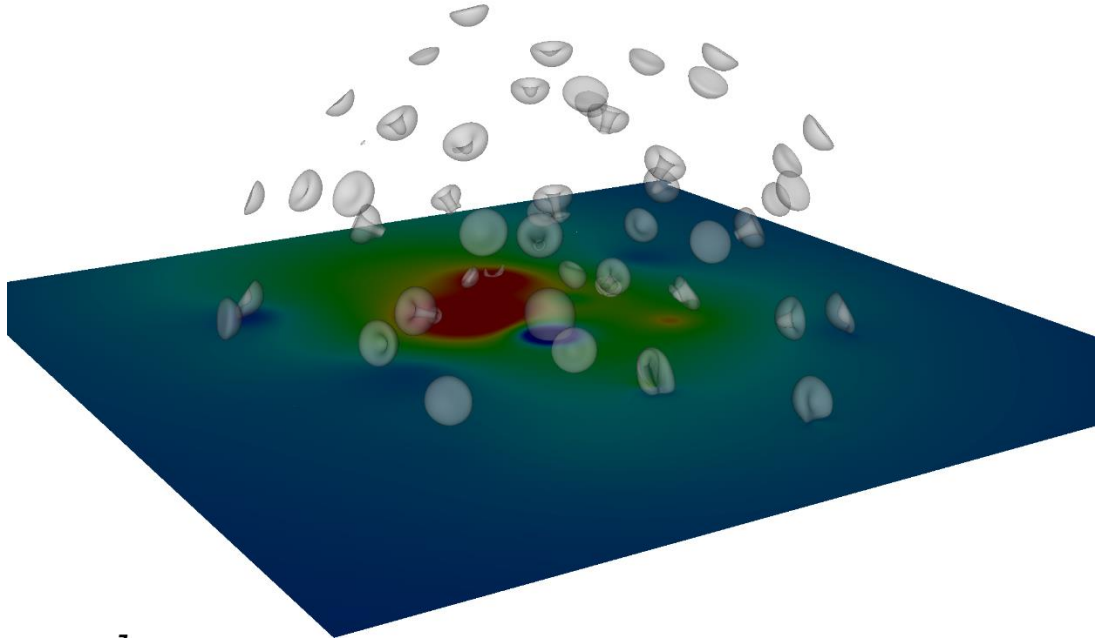
Growth and collapse of a bubble cluster

- Bubbles shown in light gray
- Coloured contours show pressure on the wall
- Bubbles undergo extreme deformation (large shape and size variations) including merger & breakup



Motivation





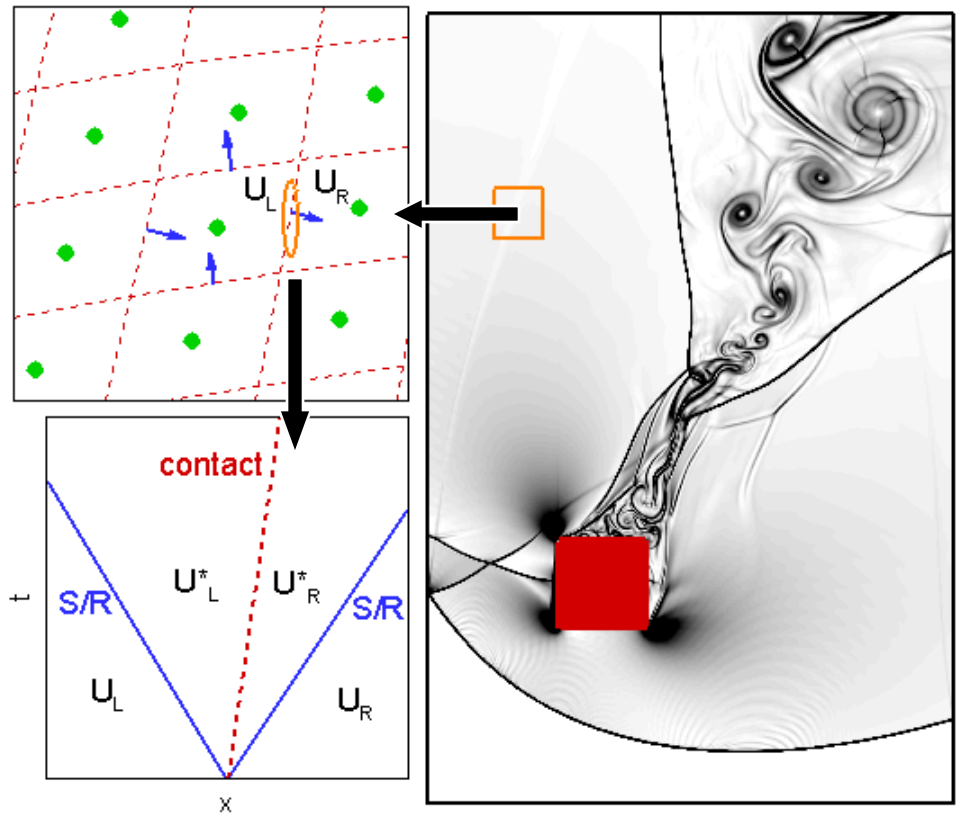
- Complex unsteady three-dimensional flow fields: complicated wave patters, interactions of shocks with material interfaces, turbulence
- Computationally challenging & expensive: Vast range of temporal and spatial length scales
- Contrasting requirements for smooth (expansion waves, vortices) and discontinuous (shocks, material interfaces) flow features
- **Need adaptive-order (multi-resolution) methods that can accurately resolve smooth as well as discontinuous flow features**

Approach

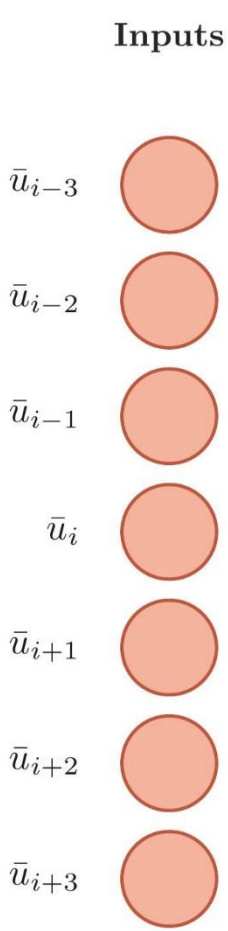
- Finite volume method for cell averaged flow variables
  - Integral form of governing equations (hyperbolic conservation law)
  - Discontinuities spread over a few grid cells
- Reconstruct and evolve
  - Use cell averaged values to compute face centered ones (Reconstruction)
  - Compute intercell numerical flux from the solution to the local normal 1D Riemann problem

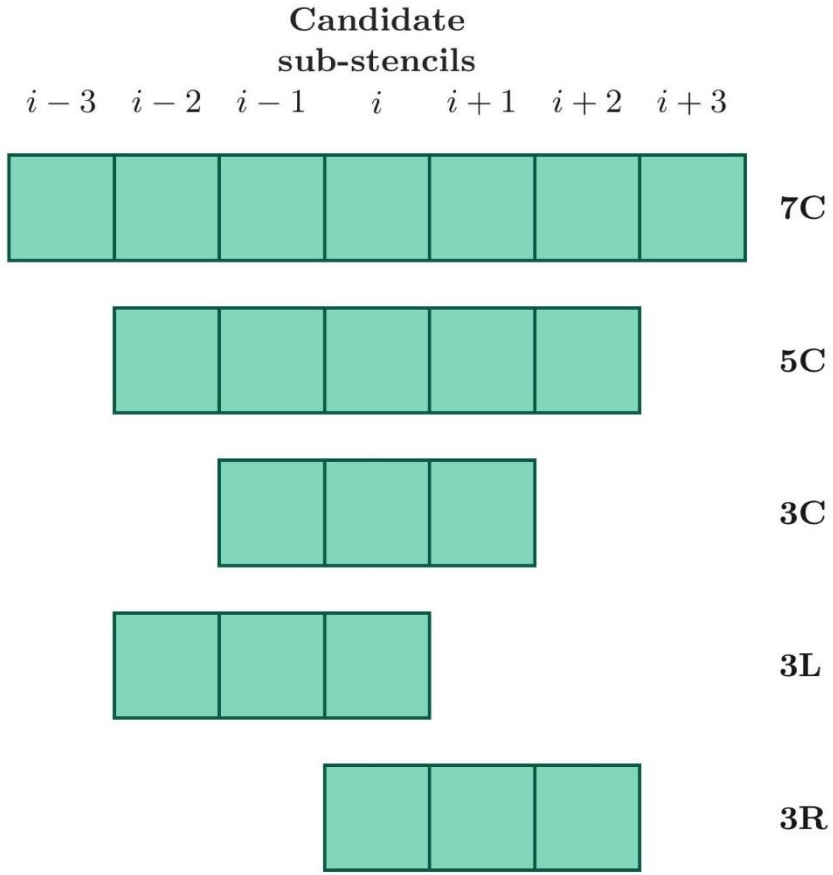
Reconstruction operation critical accuracy/resolution determining step

- Robust approximations compromise accuracy
- High-resolution methods are prone to oscillations

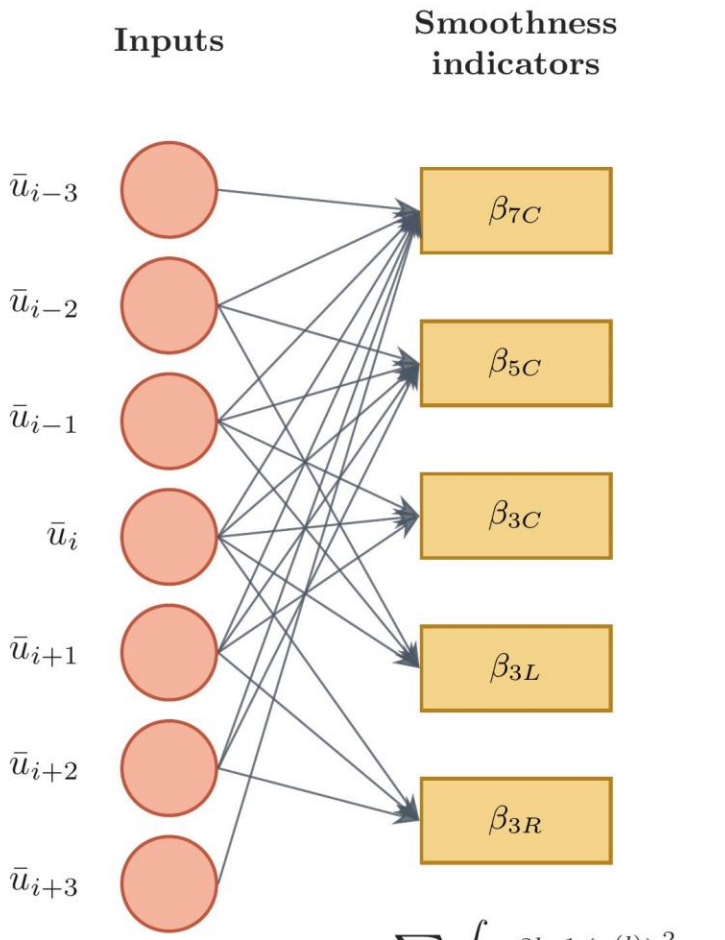


Classical high-resolution reconstruction



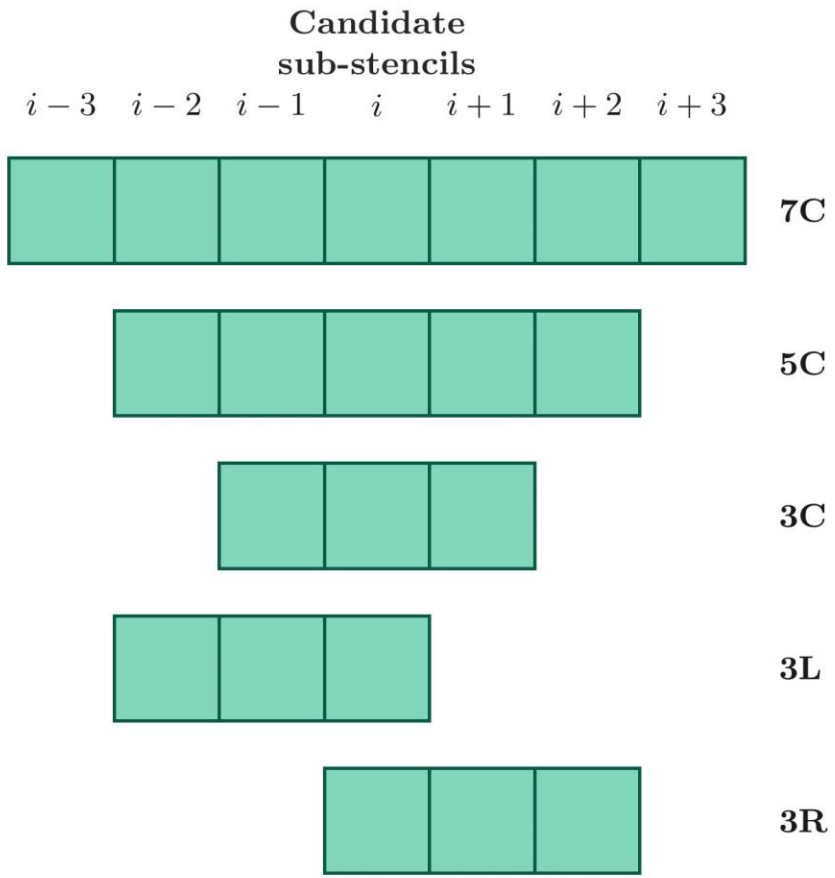


Classical high-resolution reconstruction

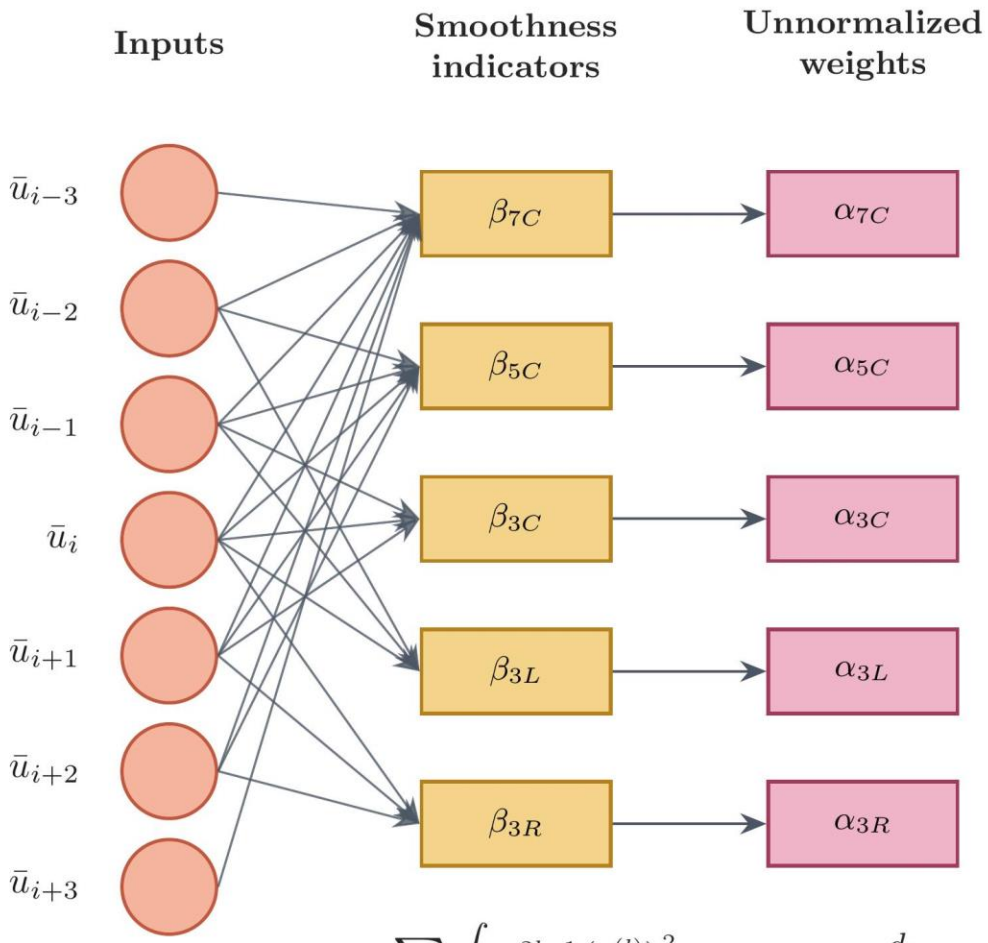


$$\beta_k = \sum_l \int h^{2l-1} \left(p_k^{(l)}\right)^2 dx$$

p_k: stencil polynomial from Taylor-series approximation

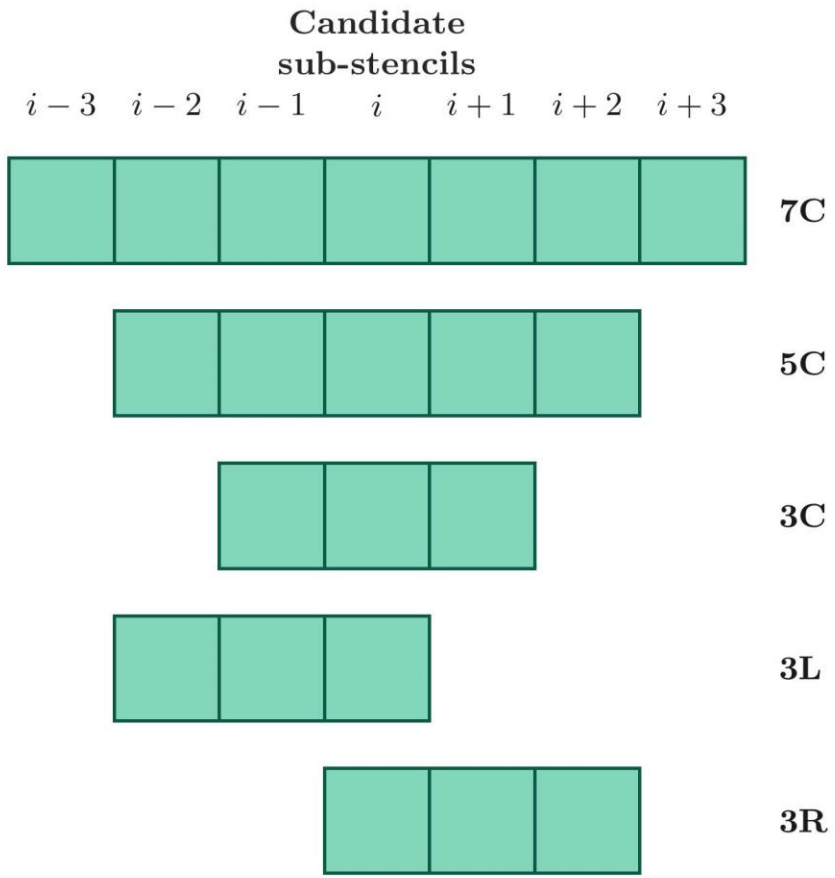


Classical high-resolution reconstruction

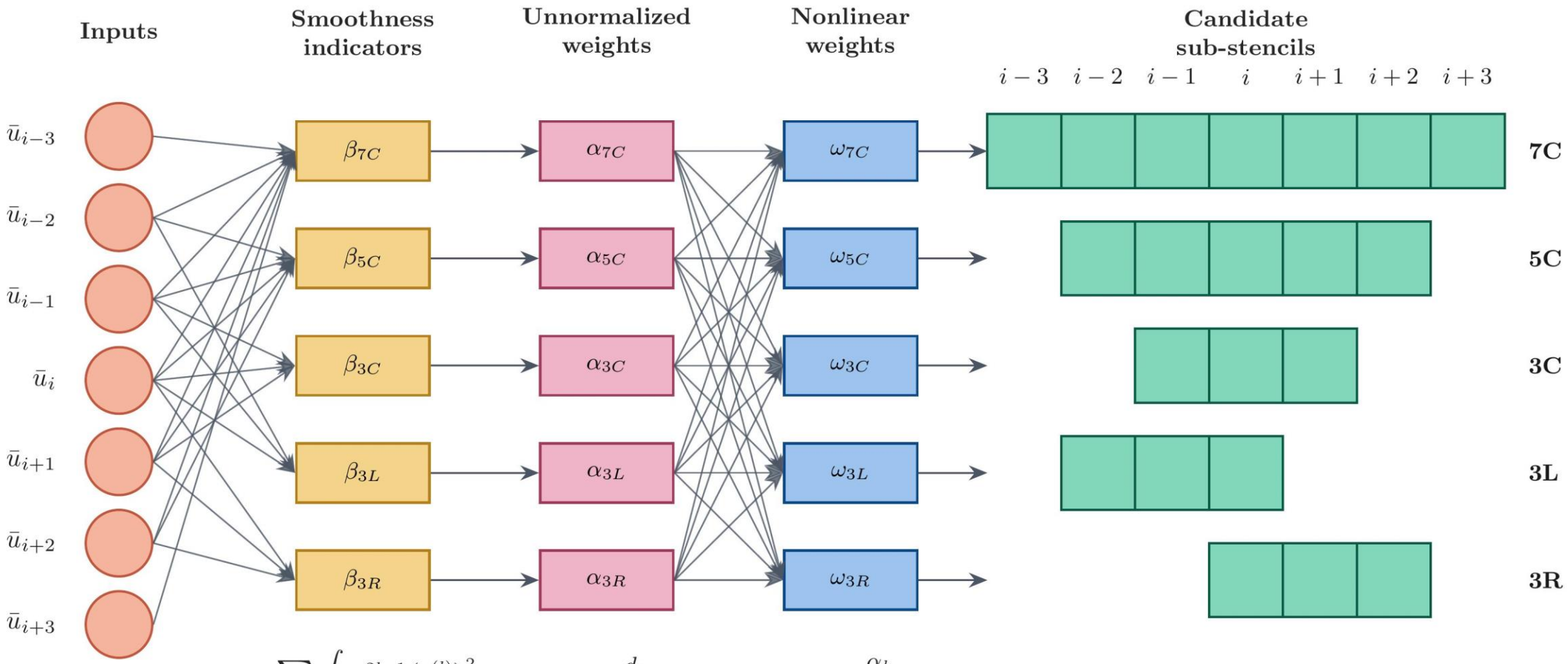


$$\beta_k = \sum_l \int h^{2l-1} \left(p_k^{(l)}\right)^2 dx \quad \alpha_k = \frac{d_k}{(\epsilon + \beta_k)^q}$$

p_k: stencil polynomial from Taylor-series approximation



Classical high-resolution reconstruction



$$\beta_k = \sum_l \int h^{2l-1} \left(p_k^{(l)}\right)^2 dx \qquad \alpha_k = \frac{d_k}{(\epsilon + \beta_k)^q} \qquad \omega_k = \frac{\alpha_k}{\sum_l \alpha_l}$$

Reconstruction $= \sum_k \omega_k \, p_k(x)$
(weighted sum of all candidates)

p_k: *stencil polynomial from Taylor-series approximation*

Classical high-resolution reconstruction

Computes several low-order approximations and then suitably combines to achieve multi-resolution reconstruction depending on the regularity of the underlying field

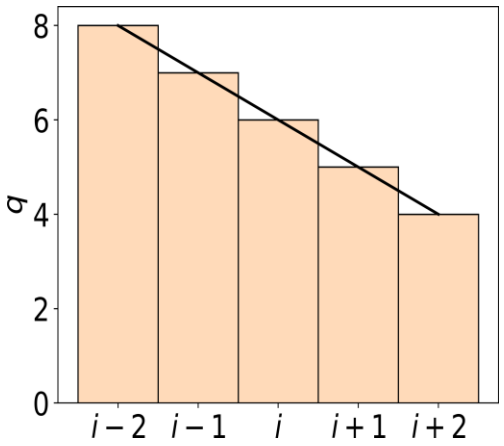
Regularity/smoothness not known a priori

Is deduced via differencing schemes constructed using Taylor Series expansions



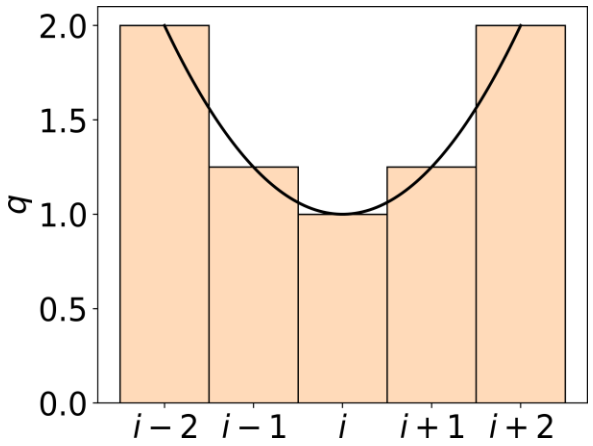
Stencil Selection





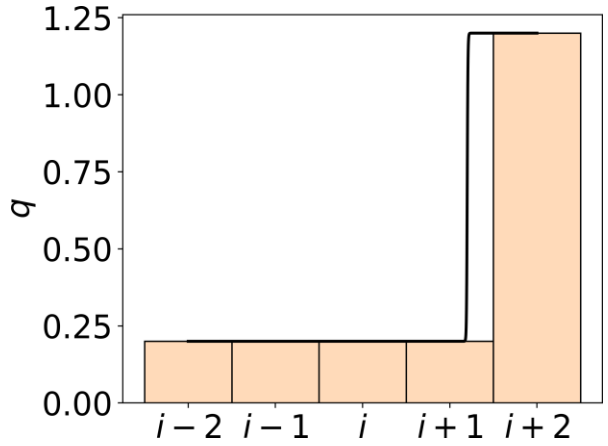
5th Order Stencil





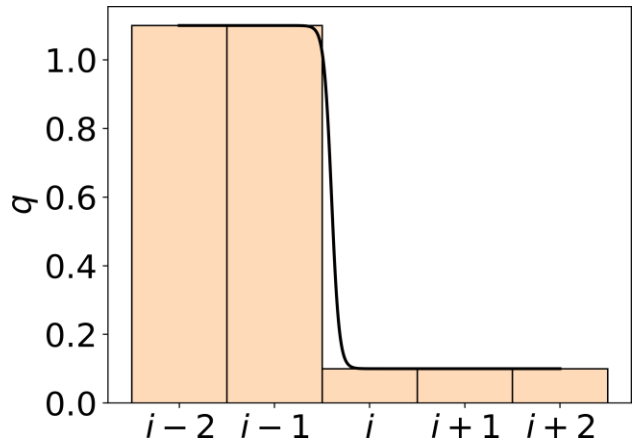
5th Order Stencil





3rd Order Central Stencil





3rd Order Right Biased Stencil

High Non-Linear Weights assigned to specific stencils
(5th order for the left two and 3rd order central/right-biased for the right two

Sensitivity w.r.t smoothness indicator

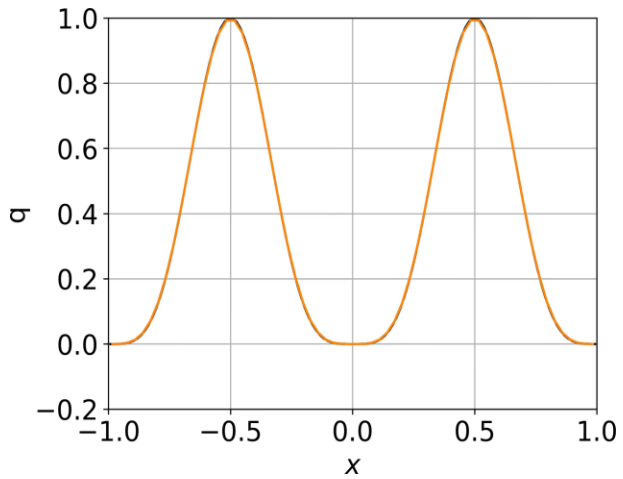
Original WENO (WENO-JS)

$$\omega_r = \frac{\gamma_r}{(\beta_r + \epsilon)^2}$$

WENO - Z

$$\omega_r = \gamma_r \left(1 + \frac{\tau^2}{(\beta_r + \epsilon)^2}\right)$$

Long time advection of a waveform (h = λ/32)



$$q_t + cq_x = 0$$

$$q(x,t) = q_0(x - ct) = \sin^4\left(\pi(x - ct)\right)$$

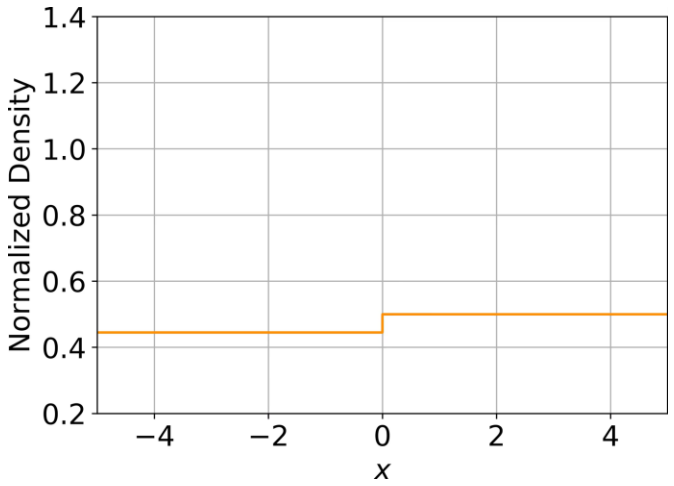
WENO - JS

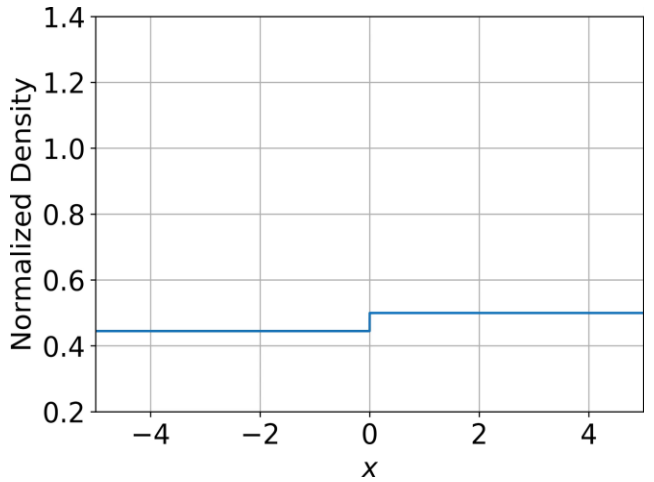
WENO - Z

Exact Solution

Lax Shock-Tube Problem

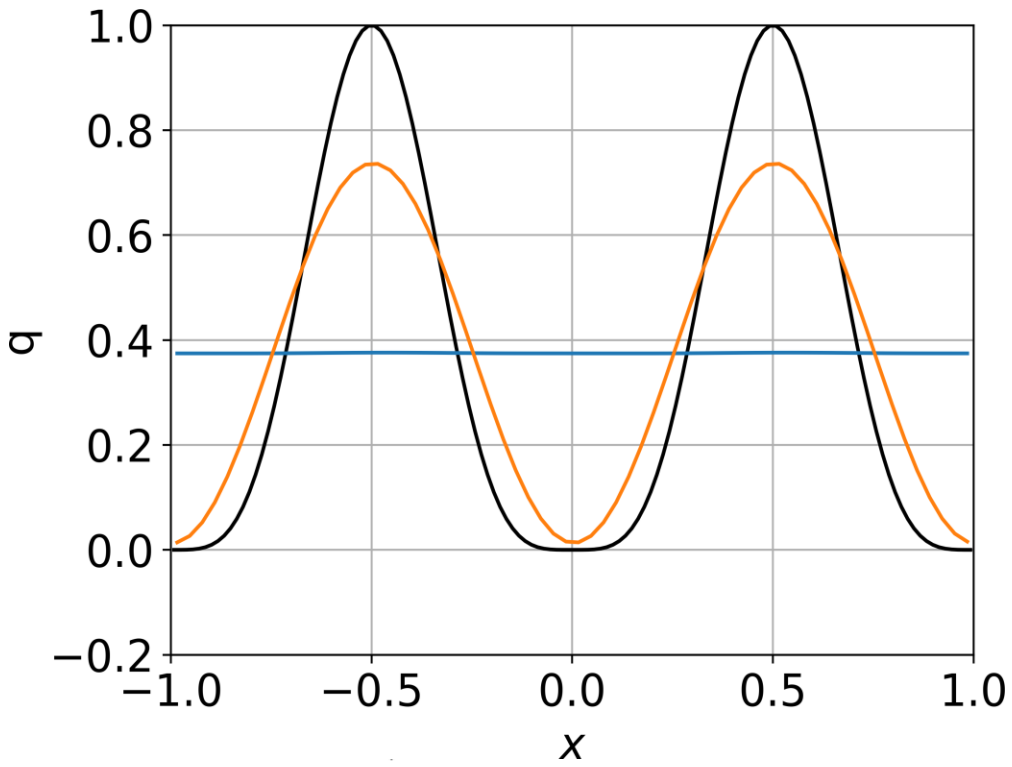
$$(\rho, u, v, p) = \begin{cases} (0.445, 0.698, 0, 3.528), & x < 0.5 \\ (0.5, 0, 0, 0.571), & x \geq 0.5 \end{cases}$$





Sensitivity w.r.t smoothness indicator

Long time advection of a waveform

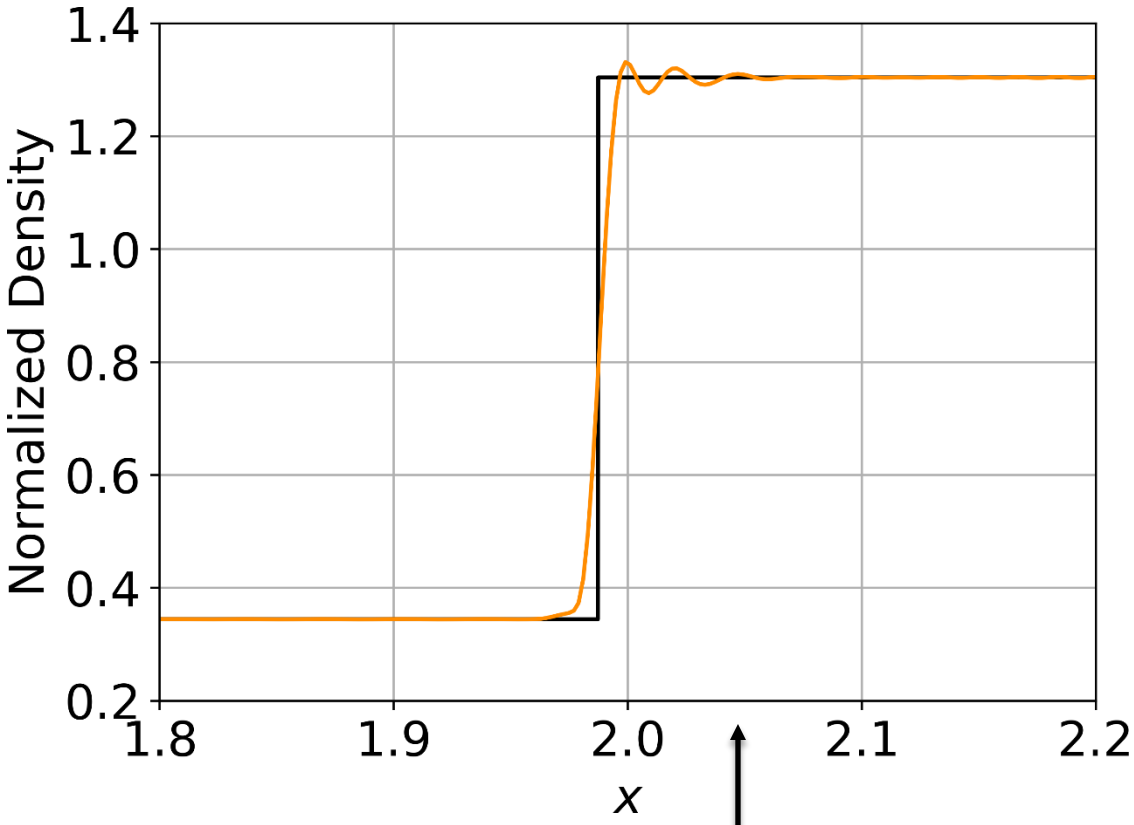


WENO - JS

WENO - Z

Exact Solution

Lax Shock-Tube Problem



Oscillations at discontinuity

Classical high-resolution reconstruction

Computes several low-order approximations and then suitably combines to achieve multi-resolution reconstruction depending on the regularity of the underlying field

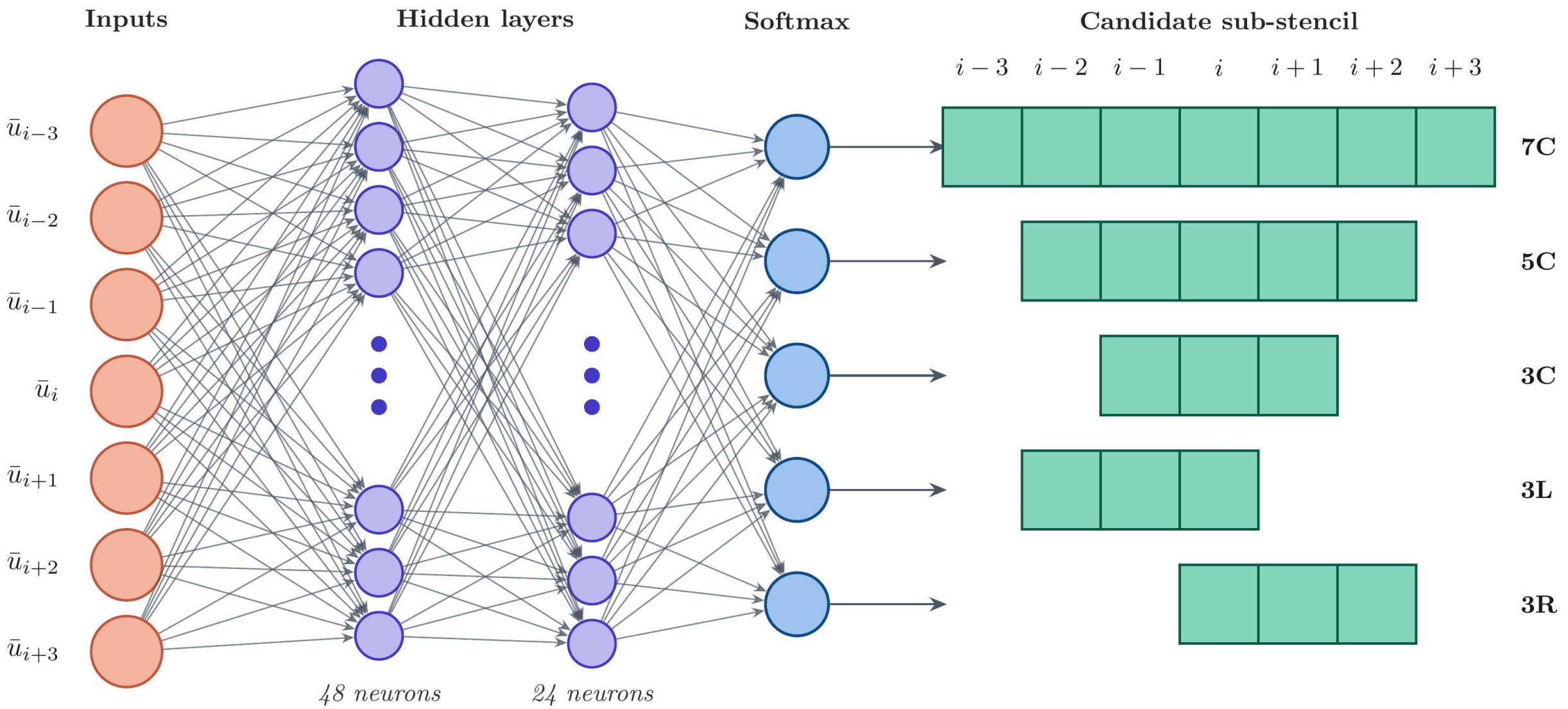
Regularity/smoothness not known a priori

Is deduced via differencing schemes constructed using Taylor Series expansions

Smoothness indicators can be quite inaccurate for marginally/poorly resolved flow features



Stencil Selection: ANN approach

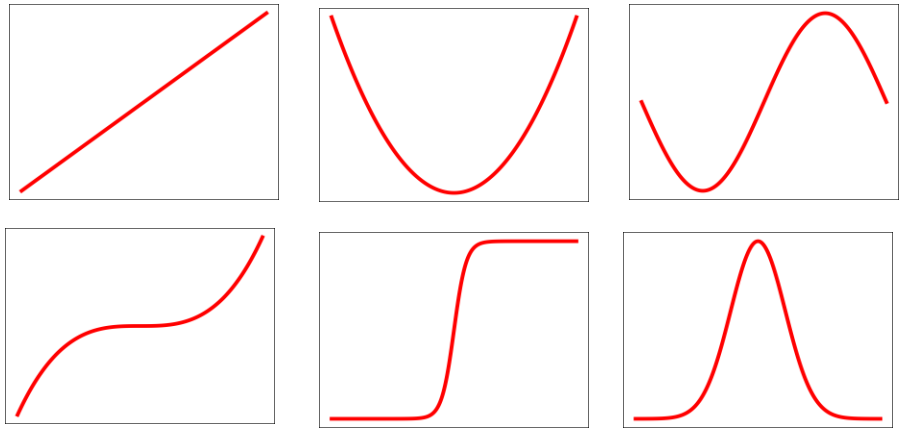


Stencil Selection: ANN approach

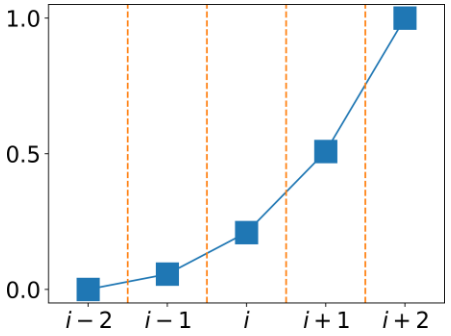
- Pose stencil/order selection as a multiclass classification problem
- Directly select appropriate stencil for reconstruction based on the accuracy
- Obviates the need for estimation of solution regularity using smoothness indicators and non-linear weights
- Exploits representational power of neural nets



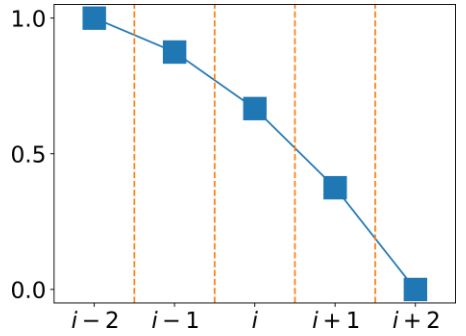
Training dataset for the network (smooth functions)



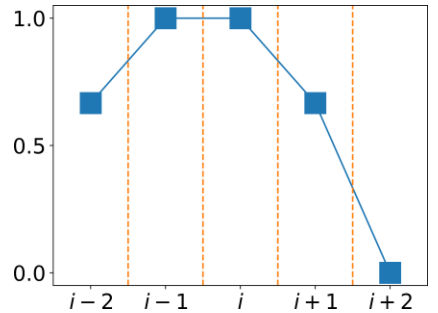
Include canonical functions frequently encountered in numerical solutions of hyperbolic PDEs



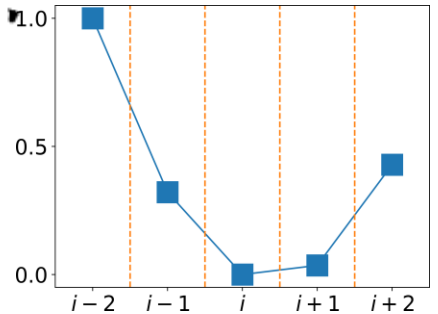
Increasing Function



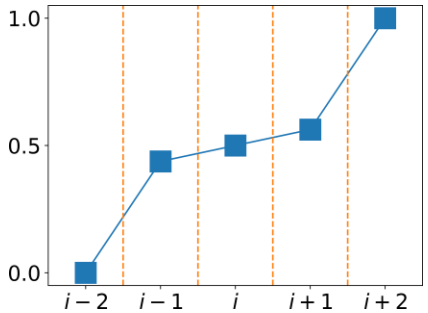
Decreasing Function



Local Maxima



Local Minima



Inflection Point

For each cell, the **lowest order stencil** (first, third central or fifth) that yields **minimum error** in the reconstructed value is chosen to be the **output label**

- **Constant function – 1st order**
- **Linear, quadratic functions – 3rd order**
- **Other functions – 5th order & 7th order**

A range of smooth functions are incorporated in the training data

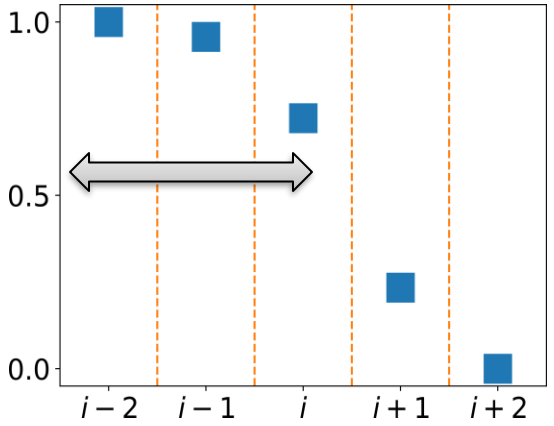
Training dataset for the network (discontinuous functions)

Location of discontinuity

Direction of discontinuity

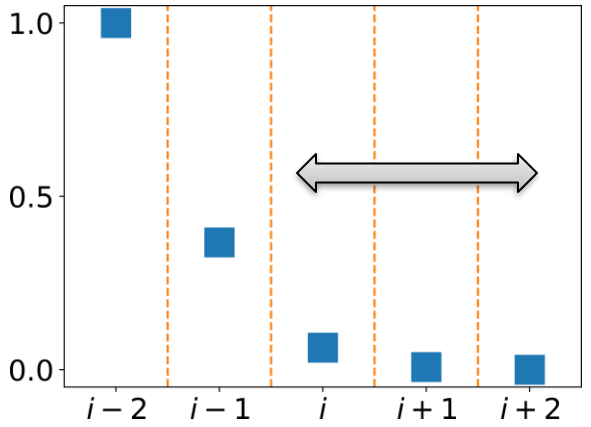
$$\frac{1}{2}\left(1 + \alpha \tan\left(\frac{x - x_0}{\epsilon}\right)\right)$$

Spread of discontinuity



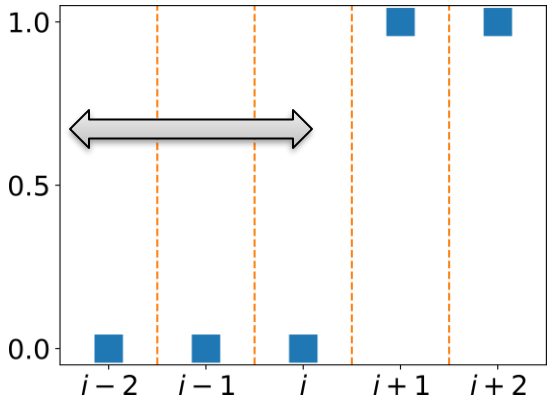
α = 1, ε = 1, x_0 = 0.5

Label: 3rd Order Left Biased Stencil



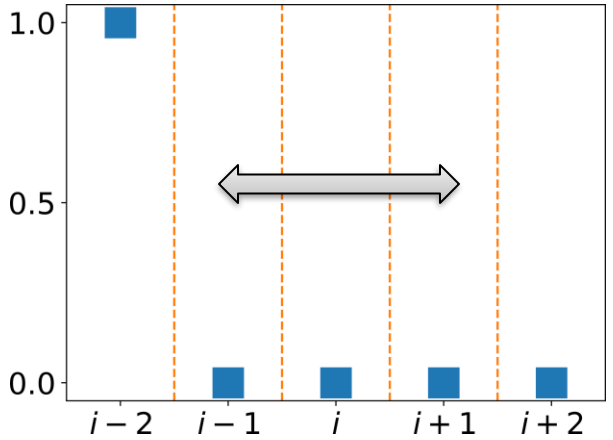
α = -1, ε = 1, x_0 = -1.5

Label: 3rd Order Right Biased Stencil



α = 1, ε = 0.1, x_0 = 0.5

Label: 3rd Order Left Biased Stencil



α = -1, ε = 0.1, x_0 = -1.5

Label: 3rd Order Central Stencil

Demonstrations

- Computations utilize a uniform mesh
- Temporal integration using 4th order 5 stage SSPRK method
- Dimension-by-dimension reconstruction for multidimensional setups

$$\frac{\partial \mathbf{Q}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{Q}) = 0$$

Linear-Convection Equation

$$\mathbf{Q} = [q], \qquad \mathbf{F} = [q\mathbf{I}]$$

Burgers' Equation

$$\mathbf{Q} = [q], \qquad \mathbf{F} = \left[\frac{q^2}{2}\mathbf{I}\right]$$

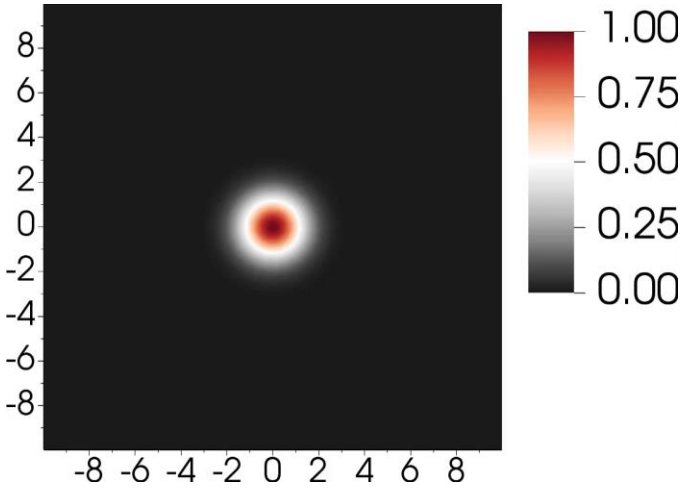
Euler Equations

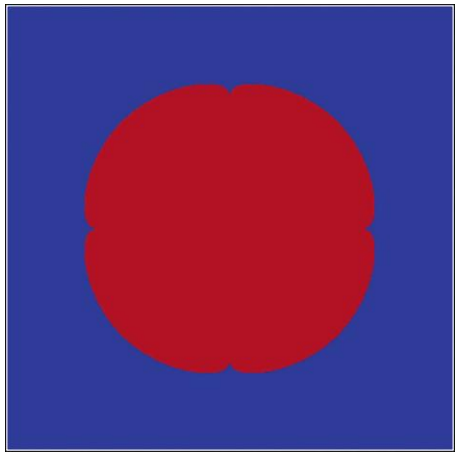
$$\mathbf{Q} = [\rho, \rho\mathbf{u}, E], \qquad \mathbf{F} = [\rho\mathbf{u}, \rho\mathbf{u}\otimes\mathbf{u} + p\mathbf{I}, \mathbf{u}(E+p)]$$

SSPRK – Strong Stability Preserving Runge-Kutta

Linear-Convection (Convergence Test)

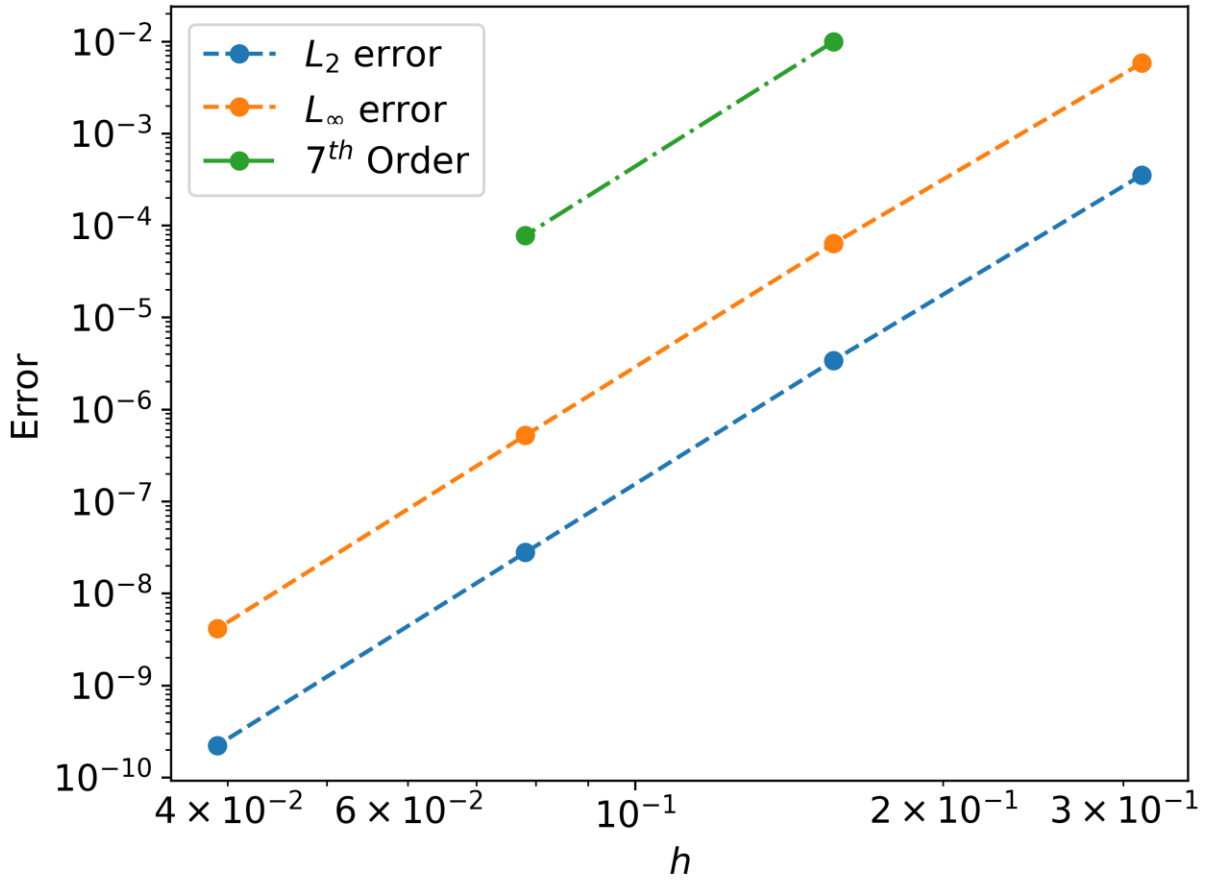
Gaussian Profile Advection $\mathbf{u} = (1,1)$





1st Order

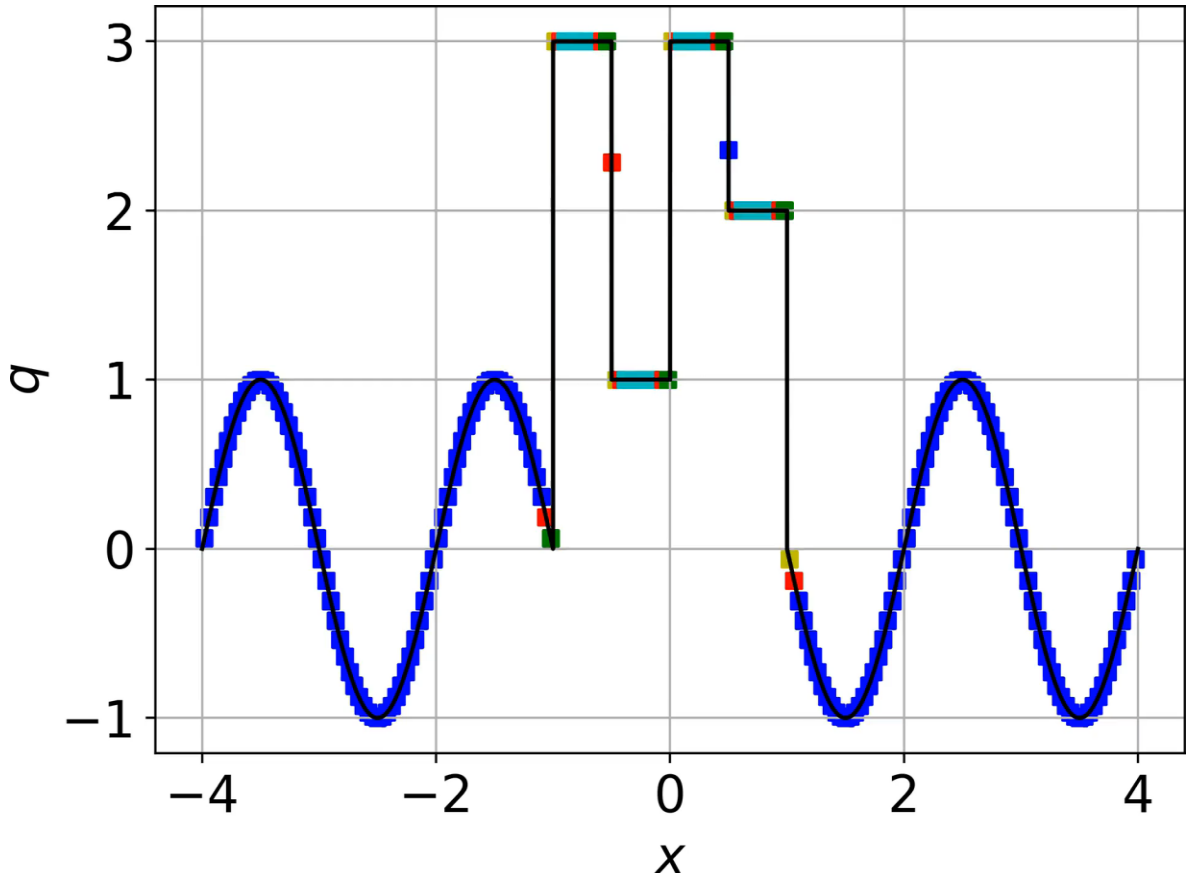
7th Order



Seventh, fifth, third and first order stencils are correctly identified

Burgers' Equation – Compound Wave*

- **Domain**: [-4,4] discretized using 200 cells
- Periodic boundary conditions
- Alternating shock and rarefaction waves develop over time



1st Order

3rd Order (right biased)

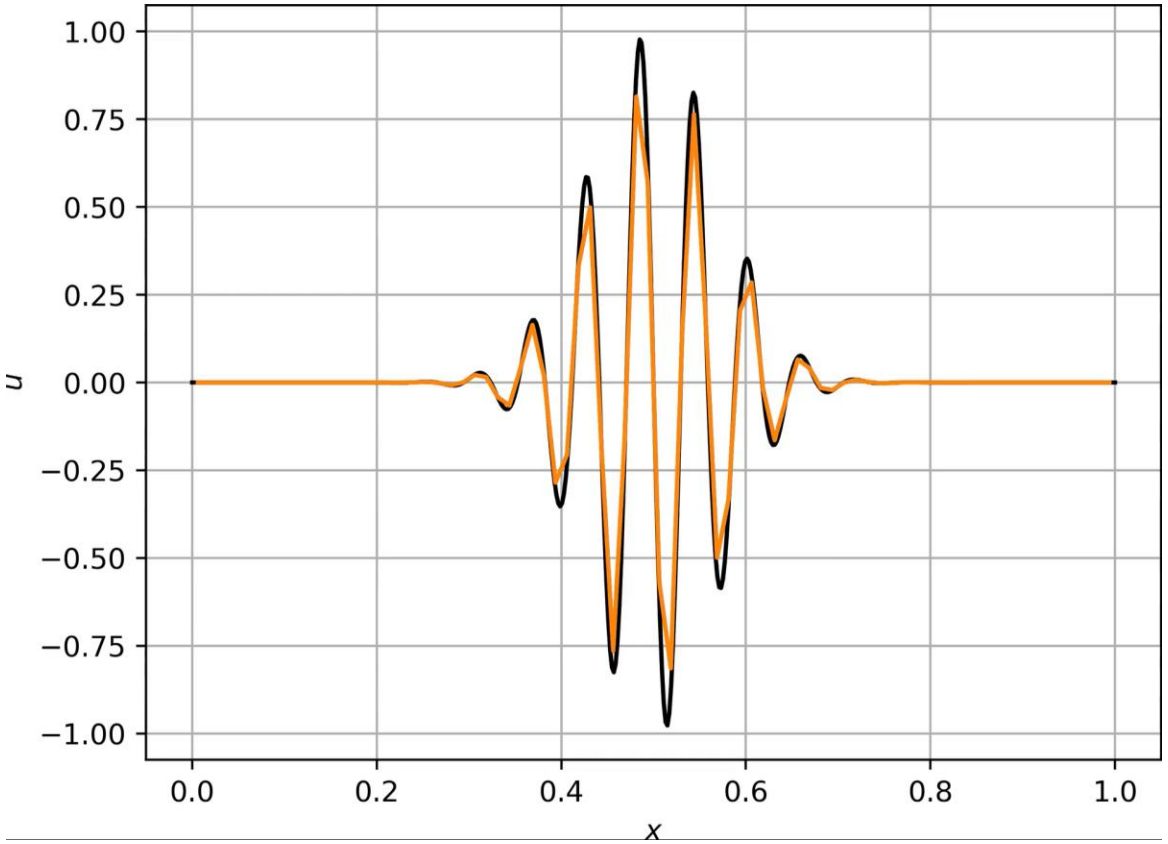
3rd Order (left biased)

3rd Order (central)

5th Order

*Ray & Hesthaven - An artificial neural network as a troubled-cell indicator, JCP 2018

Advection Equation – Modulated sine



WENO-AO (7,5,3)

ANN-AO (7,5,3)

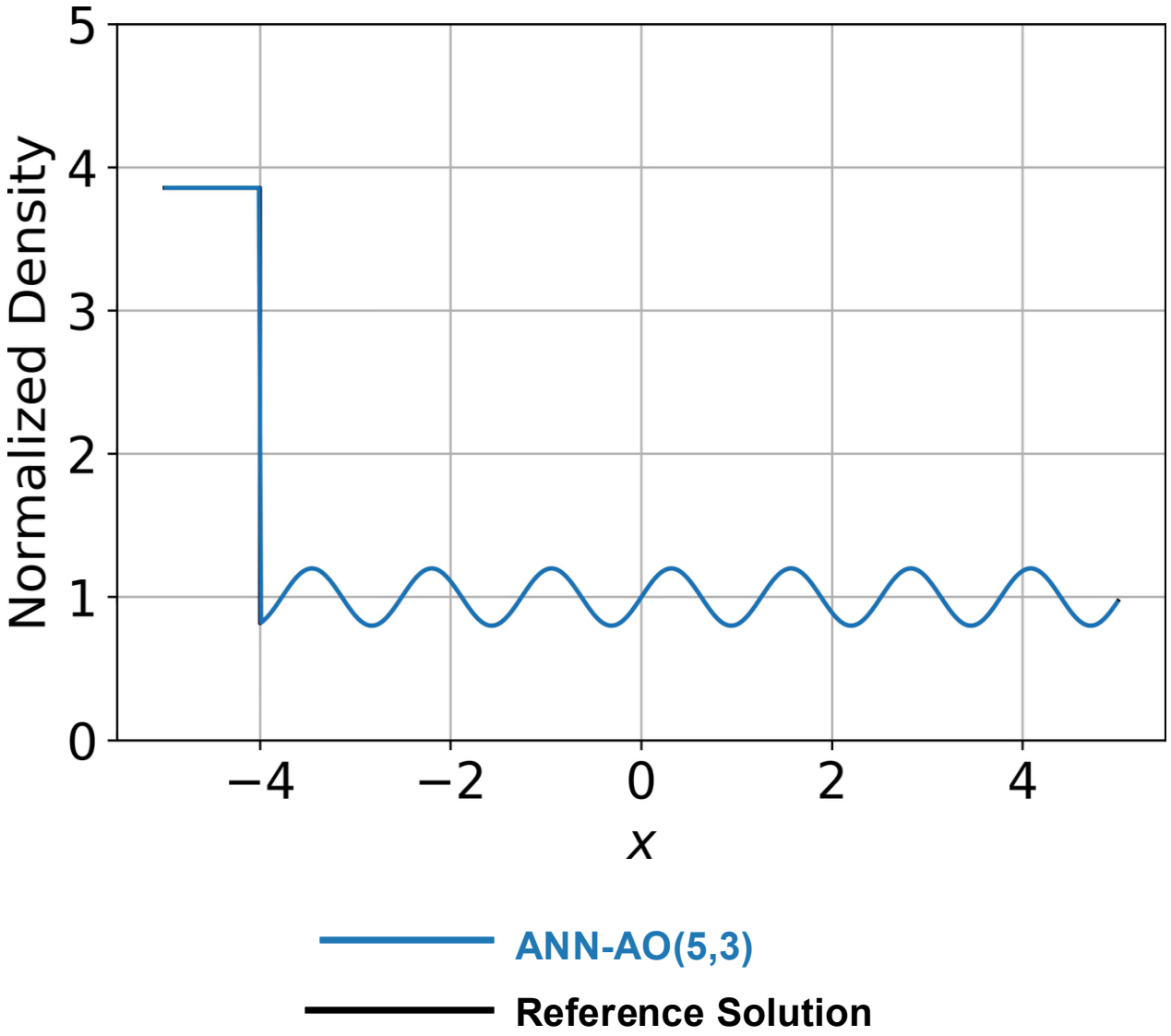
- **Domain**: [0,1] discretized using 80 cells
- Periodic boundary conditions
- The wave form is advected five times through the domain

$$q(x,0) = \exp\left(-100\left(x-\frac{1}{2}\right)^2\right)\sin(34\pi x)$$

Euler Equations - Shu-Osher Entropy Test*

- Domain: [-5,5] discretized using 400 cells
- Mach 3 shock wave interacts with a sinusoidal density perturbation
- Solution is a combination of discontinuities and smooth structures

$$(\rho, u, v, p) = \begin{cases} (3.85, 2.63, 0, 10.33), & x < -4 \\ (1 + 0.2\ \sin(5x), 0, 0, 1), & x \geq -4 \end{cases}$$

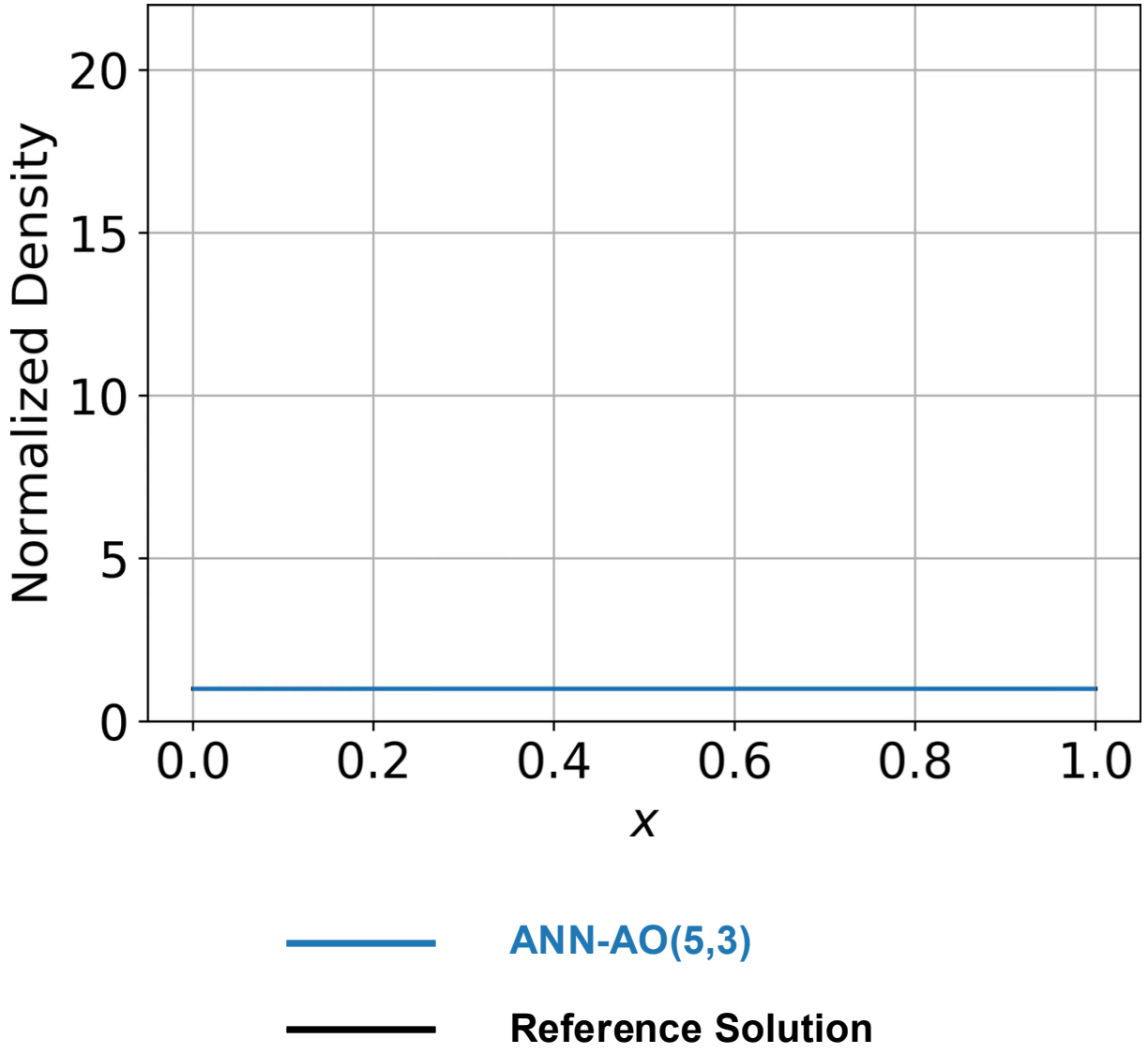


*Shu & Osher - Efficient implementation of essentially non-oscillatory shock-capturing schemes, JCP 1989

Euler Equations - Interacting Blast Waves*

- Domain: [0,1] discretized using 400 cells in confined domain
- Two strong blast waves interact in a confined domain
- Numerous interactions of discontinuities and strong nonlinear continuous waves

$$(\rho, u, v, p) = \begin{cases} (1, 0, 0, 1000), & x \leq 0.1 \\ (1, 0, 0, 0.01), & 0.1 < x \leq 0.9 \\ (1, 0, 0, 100), & \text{otherwise} \end{cases}$$

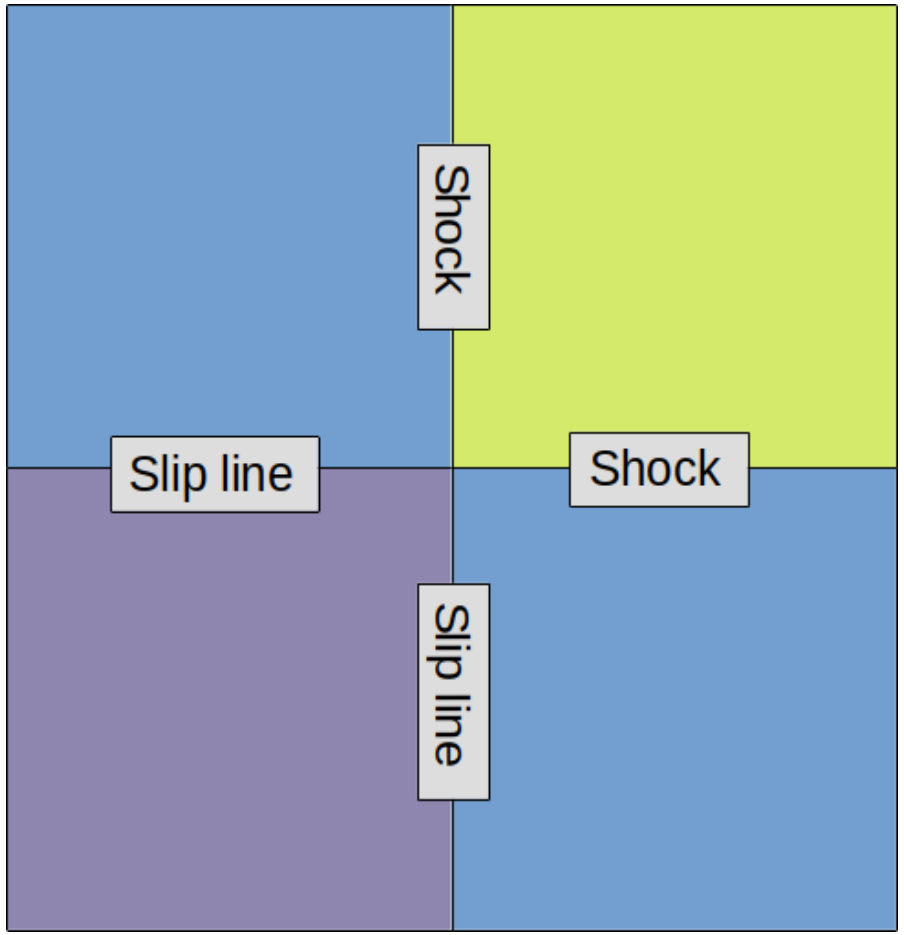


*Woodward & Collela - The numerical simulation of two-dimensional fluid flow with strong shocks, JCP 1983

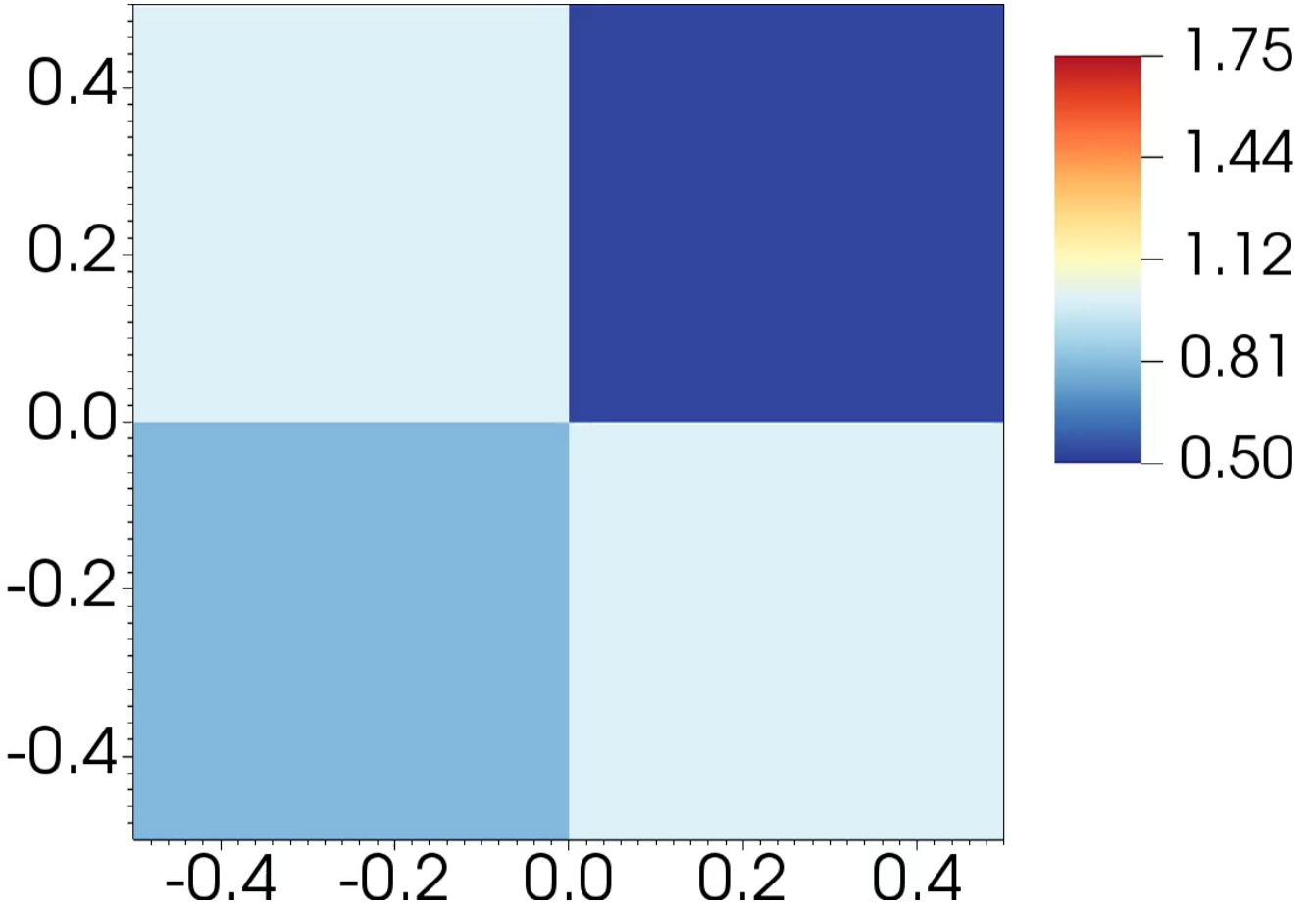
Euler Equations - 2D Riemann Problem*

2D Riemann problem - unit square divided in four quadrants consisting of constant states **separated by shock / slip line or a rarefaction**

Initial Configuration (Configuration F*)



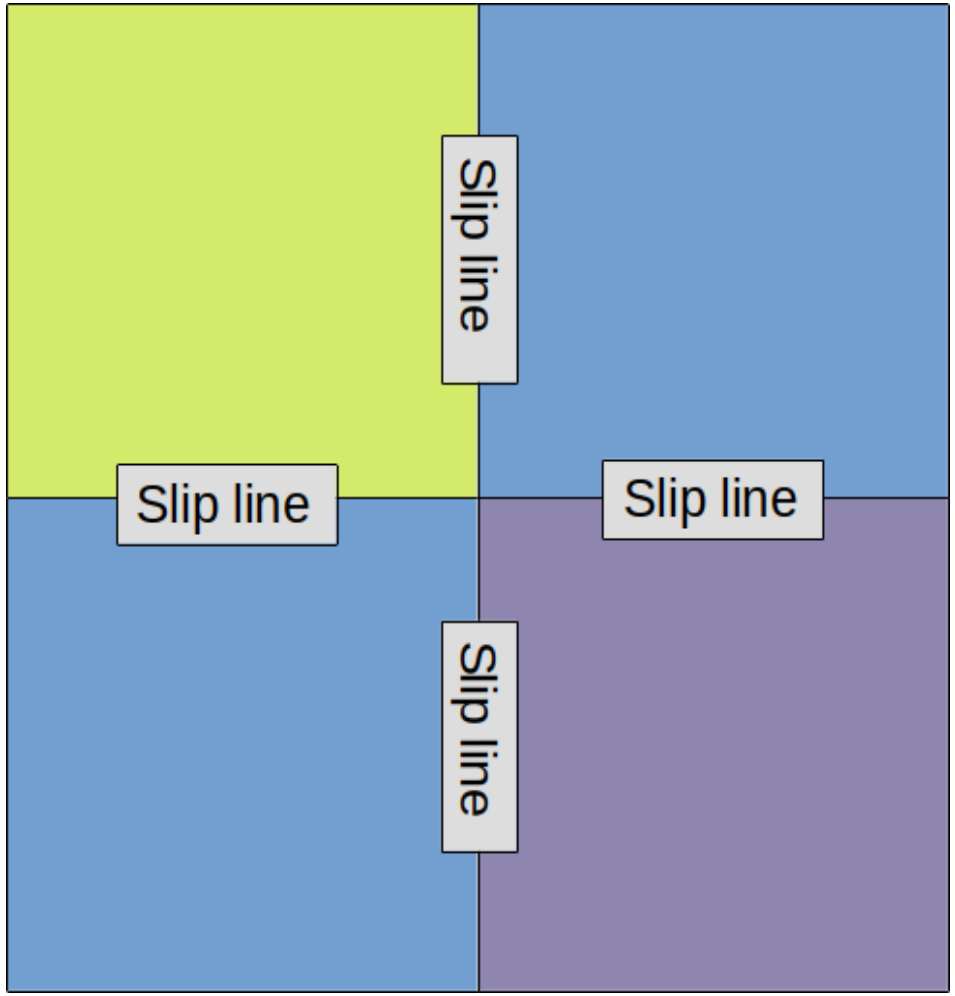
Density



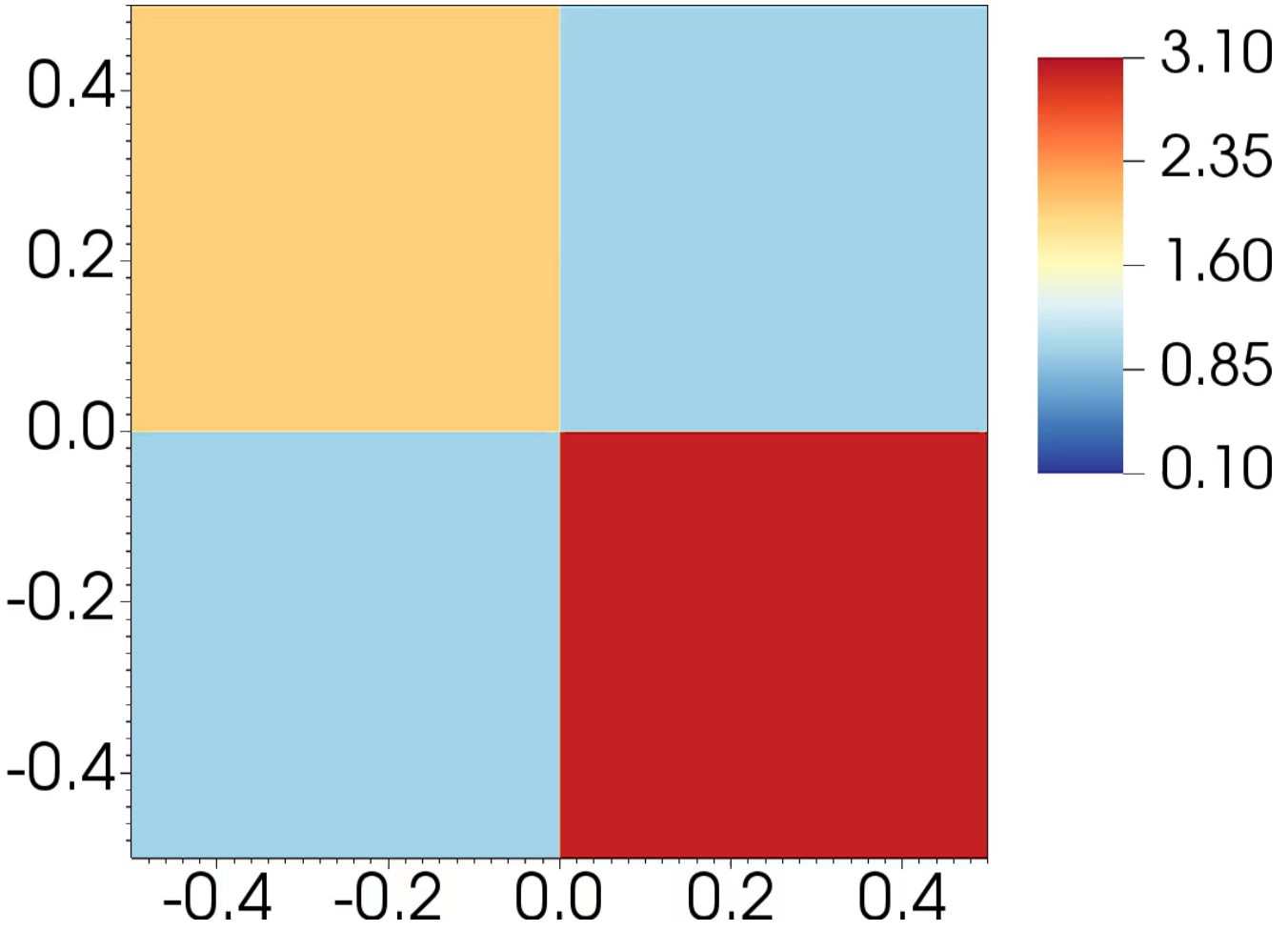
*Rinne, Collins & Glaz - Numerical solution of the Riemann problem for two-dimensional gas dynamics, SIAM JSC 1993

Euler Equations - 2D Riemann Problem

Initial Configuration (*Configuration B)

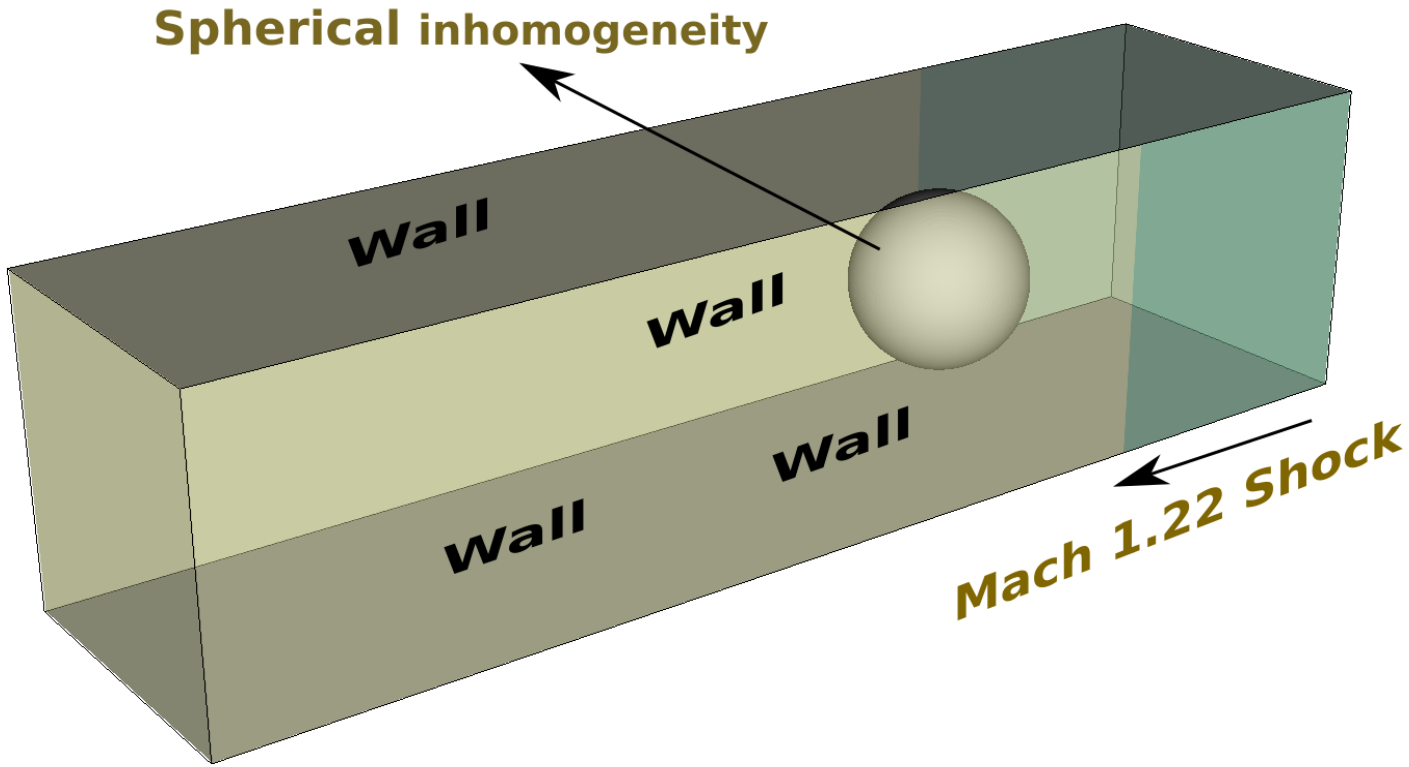


Density



Shock-bubble interaction*

Mach 1.22 shock in air impacts a spherical bubble (density ~5.5 times lower, corresponds to He)

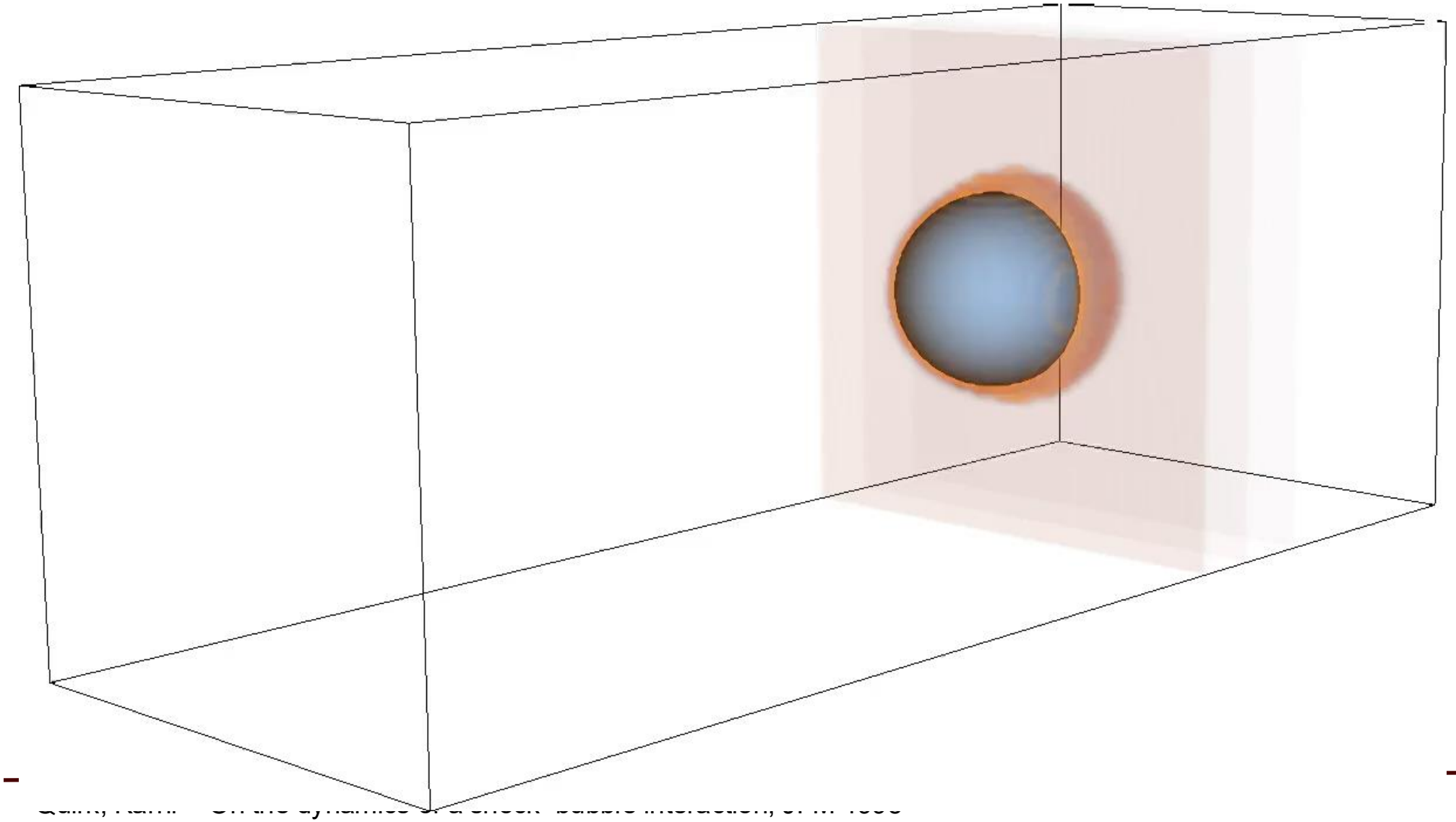




*Quirk, Karni - On the dynamics of a shock–bubble interaction, JFM 1995
*Haas and Sturtevant - Interaction of weak shock waves with cylindrical and spherical gas inhomogeneities – JFM 1986

Shock-bubble interaction*: ANN-AO(7,5,3)

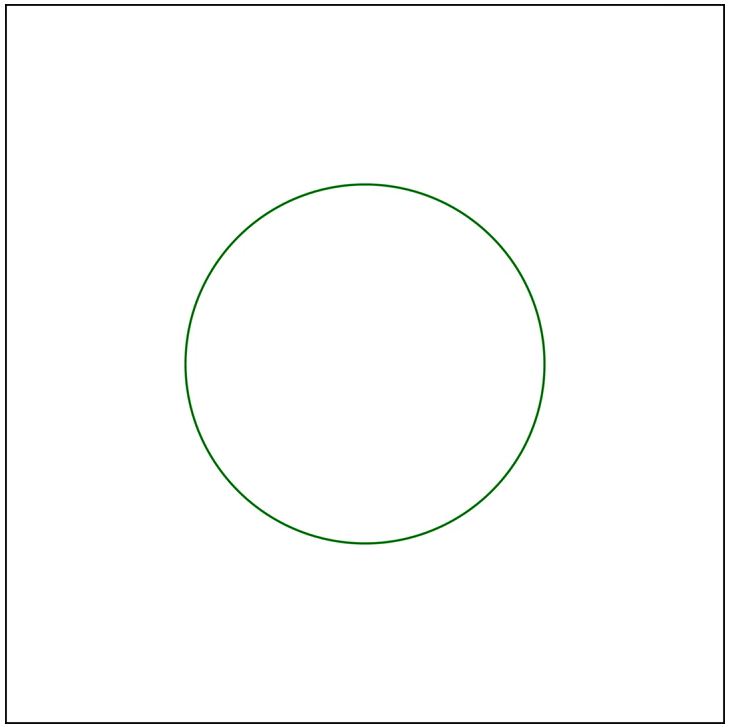
Mach 1.22 shock in air impacts a spherical bubble (density ~5.5 times lower, corresponds to He)



*Haas and Sturtevant - Interaction of weak shock waves with cylindrical and spherical gas inhomogeneities – JFM 1986

Summary

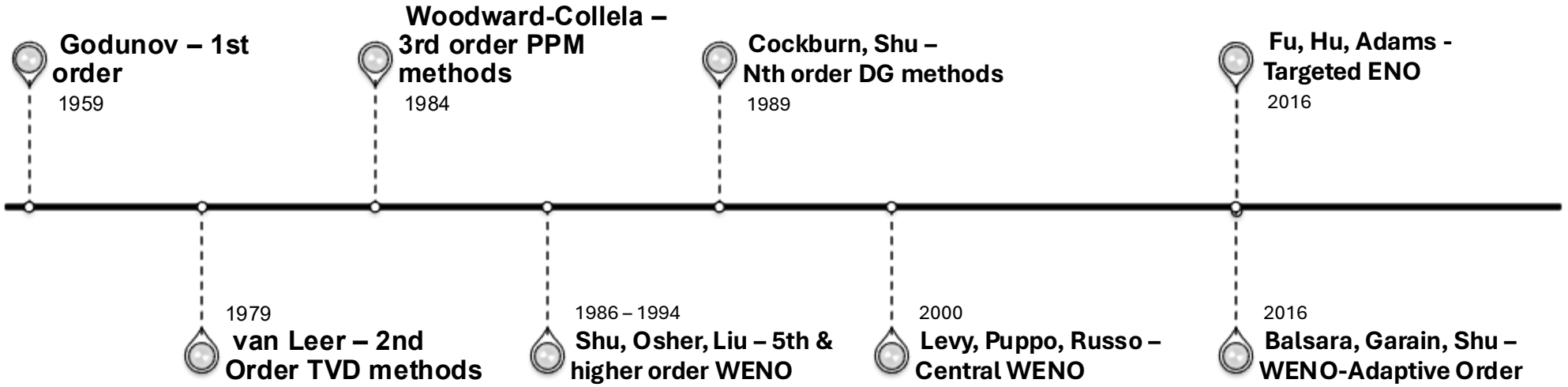
- Data-driven stencil selection & order detection method for adaptive-order discontinuity-capturing finite-volume method.
  - ANN based stencil selection and order detection
  - Obviates the need for smoothness indicators
- Demonstrated the robustness (non-oscillatory capture of strong shocks & rarefaction waves) & accuracy (low numerical dissipation) in one-dimensional and multidimensional flow simulations
- Future work: Generalization to multiphase flows (cavitation bubble dynamics), extension to unstructured mesh methods
- Acknowledgement: National Supercomputing Mission, I-X Mini-Visit Programme



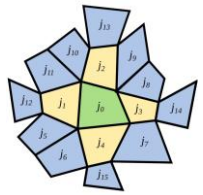
Interaction of a shock in water with an air cavity

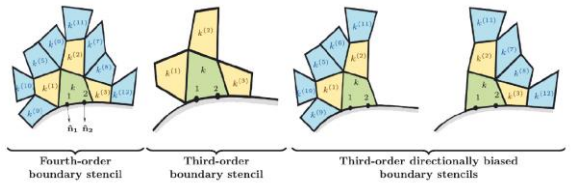
Thank you!

Adaptive-Order Methods



- ***Nominally 2nd order TVD (Total Variation Diminishing) schemes:*** a variety of limiters – Minmod, Superbee, MC, Sweby etc.
- ***WENO (Weighted Essentially Non-Oscillatory) schemes***: Modifications to the original smoothness indicators (WENO-M, WENO-Z), MP-WENO, CWENO, WENO-AO, TENO, etc.
- ***DG (Discontinuous Galerkin) Methods:*** with troubled cell indicators, WENO limiters, MOOD based limiters etc.
- CLS-WENO-AO (4,3): JCP 2022, Vol. 470, 111534





WENO Schemes

$$\max_{0 \le k \le r} \left| \frac{d^k U(x)}{dx^k} \right| \le M$$

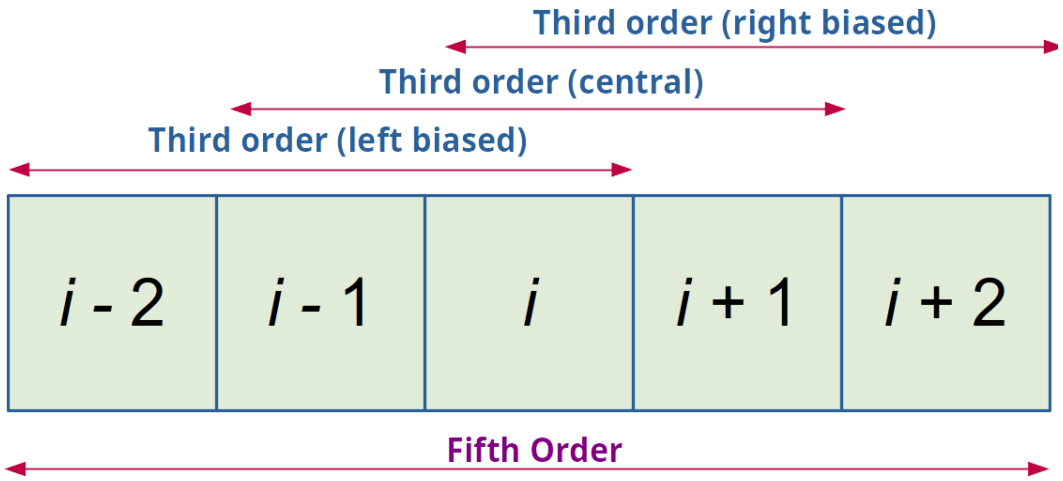
C^r class function

$$\beta_r = \sum_{l=1}^{r-1} \left(\frac{d^l U_r(x)}{dx^l} \right)^2$$

Smoothness Indicator for the ***r^th^*** order polynomial

$$\omega_r = \frac{\gamma_r}{(\beta_r + \epsilon)^p}$$

Non-Linear Weights for the ***r^th^*** order polynomial



$$\omega_r = f(\beta_r, \gamma_r, p, \epsilon)$$

Non-linear weights assigned to each stencil depend on

1. γ_r - Linear-weights assigned to the r-th order candidiate stencil
2. β_r - Smoothness indicators
3. ϵ - Small number used to avoid division by zero
4. p - Power assigned to the smoothness indicator

Multiple Objectives

- $\omega_r \to \gamma_r$ when solution is smooth
- $\omega_r \to 0$ in presence of discontinuity